A Fusion Method Using Terrestrial Laser Scanning and Unmanned Aerial Vehicle Photogrammetry for Landslide Deformation Monitoring Under Complex Terrain Conditions

Nan Jiang, Hai-Bo Li, Cong-Jiang Li, Huai-Xian Xiao, and Jia-Wen Zhou

***Abstract*— Landslides are one of the major factors threatening human life and social development. Monitoring and early warning of landslides are important parts of disaster prevention and control. However, due to complex terrain conditions, landslide monitoring in mountainous watersheds and high mountain valleys usually faces issues of high difficulty, high danger, poor data integrity, and low data accuracy compared with plains or hilly areas. Based on the fusion of terrestrial laser scanning (TLS) and unmanned aerial vehicle (UAV) photogrammetry, this article proposes a landslide deformation monitoring method that can be applied to complex terrain conditions where there are access restrictions and blind areas. The method is based on TLS to increase the number and range of ground control points (GCPs) of UAVs within the visual range and then uses assumed control points (ACPs) to reconstruct the UAV model and calculate landslide displacement in invisible areas. In addition, this article proposes and verifies a formula based on an allometricl equation to estimate the error between the calculated and real displacements based on this method. Then, we applied this method to a real landslide case and successfully realized landslide monitoring under the conditions of access limitation and blind areas. The proposed method enables landslide monitoring to overcome terrain restrictions, and it makes it possible to perform large-scale and high-precision ground deformation measurements with limited working ranges, which can be an effective tool for landslide monitoring in emergency situations and long-term risk evaluation.**

***Index Terms*— Complex terrain, data fusion, landslide monitoring, miscellaneous applications, optical data.**

I. Introduction

LANDSLIDE hazards have been a global research hotspot in recent years [1], [2]. Following the demands of economic development, an increasing number of human activities are gradually entering into high mountain valleys and mountainous watersheds to carry out large-scale projects such as mining extraction [3] and engineering construction projects, including hydroelectric [4] and railway [5] projects. However, landslides in these areas have deepened the scope and extent of their impact on human beings due to earthquakes, rainfall, and artificial disturbances [6]. In particular, landslides in complex terrain tend to be more concealed, so their early dynamic deformation processes are not easily detected [7], and their suddenness and destructiveness are usually greater than those in other regions due to the high elevation [8]. Except for the direct damage caused by landslides, landslide hazard chains, such as landslide debris flows and landslide-dammed lakes, can further lead to the amplification and continuation of disaster effects [9].

Landslide hazards are widespread and random in mountain valleys with fragile geological conditions and are usually in a constant state of development and evolution. Therefore, it is not economically feasible to manage all potential landslides with engineering measures. In addition, many early studies have shown that landslides typically experience a period of sustained deformation before their completely instability [10]. Depending on the type and mechanism of the landslide, the amount of this deformation can be as small as a few millimeters or centimeters [11]. Therefore, if high-precision landslide monitoring can be carried out, then the decision makers can judge whether the slope is in a stable state according to several millimeters or centimeters of displacement from the monitoring results, which can provide the basis for making further disaster management measures, such as strengthening monitoring, taking engineering measures, or emergency avoidance [12].

Depending on the implementation approach, landslide monitoring technologies at this stage can be broadly classified into two types: contact and noncontact technologies. Among them, contact technologies mainly include Global Navigation Satellite System (GNSS), inclinometer, and crack meter. These technologies can obtain millimeter-level surface deformation by installing sensors, power supply system, and data transmission system directly on the landslide surface [13]. However, their disadvantages are that the monitoring range and effect are limited by the number and distribution of instruments. Meanwhile, they may face great uncontrollable risks when applied to high and steep slopes or moving landslides. In con-

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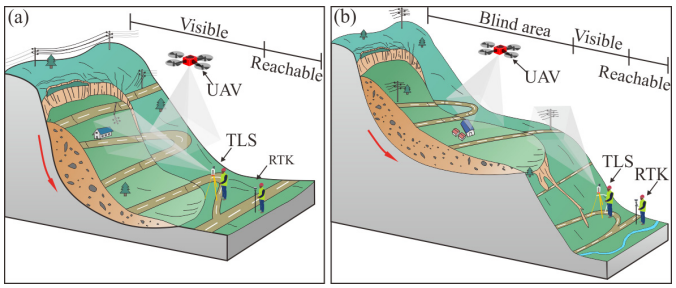


Fig. 1. Dilemma of landslide monitoring in complex terrain conditions: (a) access restriction and (b) blind areas.

trast, the noncontact landslide monitoring based on remote sensing technologies, such as unmanned aerial vehicle (UAV) photogrammetry [14] and terrestrial laser scanning (TLS) [15], can obtain the whole surface information from a distance and, thus, can reveal the overall deformation process of the landslide in a more intuitive and safer way, which is a great improvement for defining the landslide boundary, thus estimating the landslide volume and evaluating the landslide risks [16].

In particular, it should be noted that landslide monitoring in mountainous watersheds or high mountain valleys often faces the following special situations that are different from those in plain and hilly areas: access restrictions and blind areas. Access constraints can be divided into two types: terrain constraints and landslide-induced constraints. The former refers to a situation where there is no access to the landslide source in the study area, while the latter is a situation where a landslide occurs and blocks the only access road [Fig. 1(a)]. In either case, on-site monitors cannot get close to the landslide source area to carry out on-site surveys, set optical targets, or install contact monitoring equipment [17]. In addition to access restrictions, another issue with landslide monitoring in complex terrain is blind area [Fig. 1(b)]. Blind area refers to the inability to visually access the full landslide at the monitoring positions. It is usually due to a combination of high and steep slopes and access restrictions. In this case, in addition to the inability to install contact monitoring equipment on the landslide, the ground-based remote sensing techniques, such as TLS and total stations, cannot effectively monitor the landslide due to visual obstruction [18]. Therefore, the UAVs, as an airborne remote sensing technology that can quickly acquire low-altitude and high-precision terrain data, have shown advantages over other monitoring technologies in this complex environment [19].

The UAV remote sensing system is a comprehensive platform integrating remote sensing system, GPS positioning system, and inertial measurement unit (IMU) attitude determination means [20], which has higher accuracy than traditional satellite remote sensing and also has better adaptability to terrain limitations than ground-based remote sensing technologies [21]. There are a large number of researchers who have successfully applied UAV photogrammetry to landslide monitoring [22], [21], and the majority of these findings point to the key issue that ground control points (GCPs) play a decisive role in controlling the accuracy of deformation analysis using UAV photogrammetry [23], [24]. In the absence of sufficient GCPs, even with real-time kinematic (RTK) or postprocessing kinematic (PPK) to provide aerial positions and attitudes [25], the accuracy of the UAV model is still greatly reduced to tens of centimeters or even several meters, thus leading it to be used only for qualitative analysis of landslide evolutionary states rather than for surface deformation analysis at the subpixel level (a few millimeters or centimeters). Many previous landslide monitoring studies using UAV photogrammetry were conducted under conditions where GCPs could be set manually [26], [27]. However, in areas where GCPs cannot be set up, they usually have to tolerate low precision data and then take qualitative or quantitative analysis for landslide monitoring [28], [29]. Therefore, for landslide monitoring under the access restrictions or blind areas mentioned in this article, there are still many limitations in the practical use of remote sensing technologies such as UAVs and TLS.

For landslide monitoring under access restrictions, a previous study demonstrated that the fusion of TLS and UAVs can be used to improve data integrity as well as accuracy [30]. The core of this approach is to provide remote control points and check points for UAV photogrammetry using local high-precision point clouds from TLS, and the remote control points have been proven as an alternative to manual GCPs acquired by RTK. Based on this approach, we can break the limitation of access restrictions during landslide monitoring and extend the deployment of UAV control points to all areas within the field of view. However, once the landslide is located in the blind area, the farthest remote control points can only be set around the boundary of the visual range, so the accuracy of UAVs in the blind area is still not guaranteed. We have faced these dilemmas many times while conducting landslide studies in the mountainous regions of Southwest China. And so far, there are few studies on effective UAV landslide monitoring methods for blind areas.

Encouraged by the above factors, this article proposes a method for landslide deformation monitoring applicable to complex terrain based on the integration of TLS and UAV photogrammetry. For the access restriction within the field of view, we use the abovementioned TLS and UAV fusion method for high-precision UAV model construction. Then, we propose a UAV model reconstruction process and error estimation method based on assumed control points (ACPs) to achieve surface deformation analysis within the blind areas. This reconstruction process forces the multiperiod UAV models to be distorted in the same direction and magnitude in an artificially defined manner within the blind areas, so that the irregular distortion of the UAV models due to the lack of GCPs can be eliminated and an accurate alignment between the multiperiod data can be achieved. For the quantitative analysis of displacement errors based on the method, we proposed an error estimation formula by analyzing 12 scenarios of UAV models to predict the error between the actual displacement and the calculated displacement (CD) obtained by the method, and found that the error was related to the distance to the

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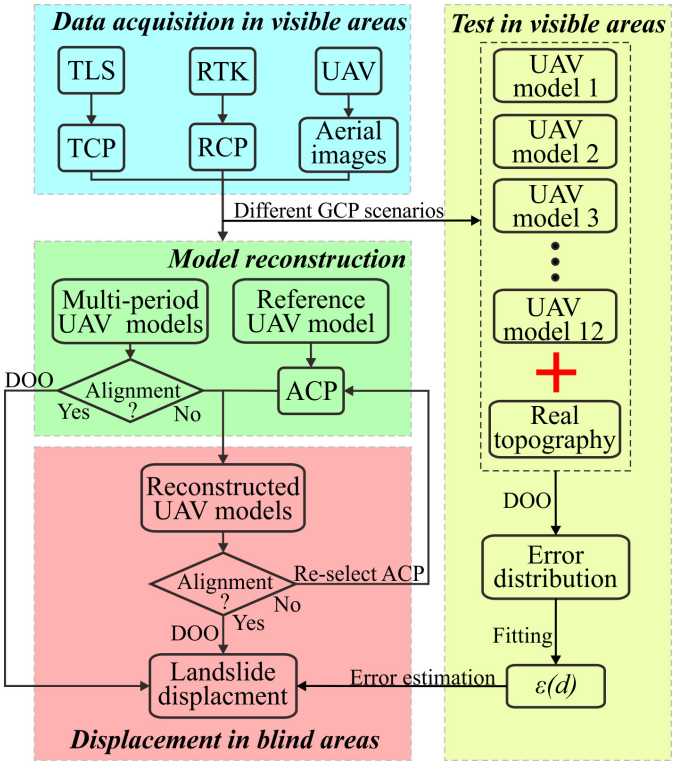


Fig. 2. Flowchart of this study: data acquisition is conducted in visible areas with TLS, RTK, and UAV; 12 groups of comparative tests were conducted to analyze the error distribution of UAV model; the ACPs are used to reconstruct the distorted UAV model to analyze the landslide displacement in blind areas.

actual GCPs: about 0.02%–0.04% for 100 m, 0.16%–0.18% for 400 m, and 0.30%–0.33% for 800 m, which is acceptable for landslide monitoring.

In general, the method proposed in this article can greatly reduce the preliminary work required in actual landslide monitoring and make surface displacement analysis using remote sensing technologies in complex environments more feasible. We prove that the error ratio of this method is very small, which can fully meet the needs of landslide deformation monitoring. Based on this approach, disaster management decision makers can obtain subpixel level surface deformation within the blind areas more easily, effectively, and safely, which is not available in the current stage of traditional methods, and thus can greatly improve the management capability of landslide disasters in mountainous areas. Meanwhile, this method can enhance the usability of remote sensing technologies, such as UAV and TLS, in surface deformation detection and can promote the optimization and innovation of remote sensing technologies in this field.

II. Materials and Methods

The materials and methods in this article mainly include on-site data acquisition in visual areas, UAV model reconstruction based on ACP, surface displacement analysis using distorted model, and derivation of the error estimation. The flowchart is shown in Fig. 2.

A. Study Area and Data Acquisition

In this study, we selected the Qingliu (QL) H02 landslides in XiaMeng (XM) township, Lixian County, located in the

TABLE I
Details of Data Acquisition

Equipment	Data acquisition	Parameters
RTK: Huace TS5 Pro	10 ground control points	Accuracy=±2 mm
UAV: Feima D2000	565 aerial images on 2020.12.06; 571 aerial images on 2021.05.10	Resolution=5 cm
TLS: Riegl VZ-2000i	Three scan positions on 2020.12.06 and 2021.05.10, respectively	Vertical and horizontal angular resolution= 0.03°

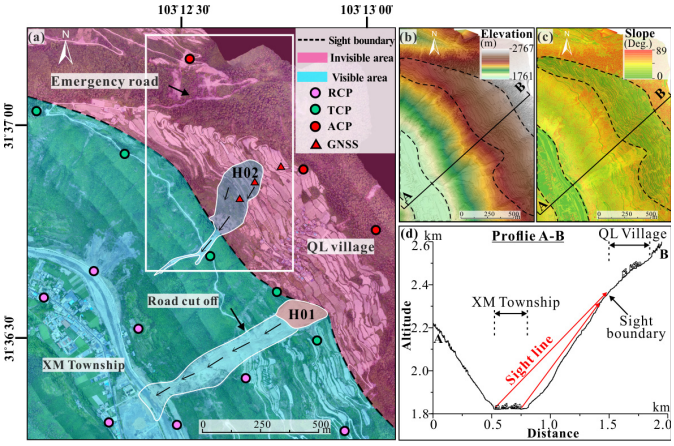


Fig. 3. Study area: (a) UAV orthophoto of the study area; the green/red indicates the visible/invisible areas from the valley after the H01 landslide; (b) and (c) elevation and slope maps of the study area and (d) typical profile of the study area, highlighting the sight boundary.

mountainous region of Southwest China, as the subject of our study [Fig. 3(a)]. QL village is located on the northeastern slope of XM township, with a relative height difference of approximately 600–800 m and a hill slope of approximately 50°–60° [Fig. 3(b) and (c)]. The QL landslides contain two independent parts, H01 and H02. H01 has been deformed significantly since August 2009, and the landslide occurred on December 4, 2020, destroying the only road leading to QL village. H02 began to show significant tensile cracks in September 2020, and the deformation continued during the postdisaster reconstruction of H01.

The initial data acquisition started with the emergency treatment of H01 on December 6, 2020. We used RTK to obtain geodetic coordinates and TLS (Riegl VZ 2000i [31]) and UAVs (Feima D2000 [32]) for terrain acquisition. The main problem with this data acquisition was the presence of access restrictions and blind areas in the H02 landslide area, and our on-site works were limited to the river valley areas. After five months, we conducted data collection again using the above techniques. The details are shown in Table I.

Due to the lack of preliminary investigation during the first period of data acquisition, information such as the scale and volume of the H02 landslide was unknown, so it was impossible to judge its movement process or potential destabilization risk at that time. Coupled with the fact that the road to QL village was damaged, an effective and comprehensive geological survey could not be conducted until the emergency road was built (approximately one month later). Since H02 happens to be located within the blind area of the field of view,

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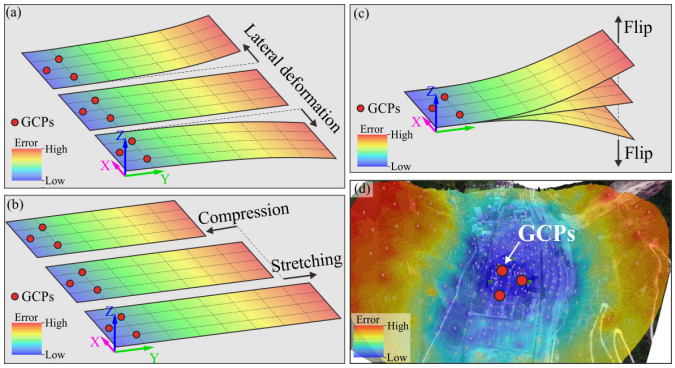


Fig. 4. Distortion of the UAV model outside the influence range of GCPs: (a) lateral deformation in the *xy* plane; (b) compression and stretching along the *xy* plane; (c) flip in vertical direction; and (d) example of UAV model distortion outside the GCPs.

when observed from the slope foot, neither manual observation nor TLS could obtain its full picture [Fig. 3(d)]. Although UAV aerial photogrammetry was used for terrain acquisition, it could not be used for accurate surface deformation analysis in the absence of GCPs.

In general, the access restrictions and blind areas of H02 led to the inability to conduct effective landslide deformation monitoring or landslide potential hazard assessment during the disaster disposal of H01. Current monitoring technologies and methods could not meet realistic needs in this scenario. This also resulted in a great uncontrollable risk for the disaster management of H01 throughout the first month.

B. Control Points for UAVs

Compared to TLS, UAV photogrammetry has a wider visual from the air; thus, it is more capable of terrain acquisition in high and steep valleys. One of the most important factors infecting the accuracy of UAV photogrammetry is the GCP. As shown in Fig. 4, many studies have pointed out that the error of the UAV model will increase with the distance from the GCPs [25], and the distortion can be generally divided into lateral deformation, compression and stretching, and flipping.

This error is caused by the cumulative error under various factors in the structure from motion (SfM) algorithm [33]. The SfM algorithm is an elaborate manual method of constructing 3-D models from 2-D images. It mainly consists of the feature extraction and matching (such as scale-invariant feature transform (SIFT) and its derivative algorithm), the geometric verification [such as random sample consensus (RANSAC)], triangulation, and bundle adjustment [34]. Among them, bundle adjustment is the key to reduce the cumulative error in the SfM algorithm [35]. It can mainly resolve the initial errors that occur during feature matching as well as triangulation. These errors come from a wide range of sources, for example: the difference in size and sharpness of the pixels may lead to errors in the projection coordinates of the matched points in the photograph; and there may be errors in estimating the essential matrix E or the fundamental matrix F; or there may be some errors in the internal parameters within the camera itself. The accumulation of these errors results in a certain distance between the reconstructed target point and the real target point, and bundle adjustment is applied to integrate the reprojection errors of all reconstruction points and then minimize these errors. It is the joint nonlinear refinement of camera parameters $\boldsymbol{P}_c$ and point parameters $\boldsymbol{X}_k$ that minimizes the reprojection error (x_j is the real position) using a function π that projects scene points to image space and a loss function ρ_j to potentially downweight outliers [36]

$$E = \sum_j \rho_j \left(\left\| \pi \left(\mathrm{P}_c, \boldsymbol{X}_k \right) - x_j \right\|_2^2 \right). \tag{1}$$

And the Levenberg–Marquardt [37] is an effective approach to solve the bundle adjustment.

In addition, the aerial images are registered to the current model one by one in the SfM process, thus carrying the remaining reprojection error from the previous bundle adjustment [38]. In the absence of more external geometric information, this cumulative error cannot be eliminated even by the global bundle adjustment after matching all the photographs [39], which is why the error of the UAVs gradually increases with the distance. For this situation, GCPs can provide additional external information about 3-D geometry, and the accuracy of the UAV model can be increased by rerunning the bundle adjustment using GCPs. Once the GCPs are available, the accuracy of the UAV model in the area enclosed by the GCPs (at least three GCPs) appears to be significantly improved, and the accuracy of the UAV model at this point depends on the accuracy of the GCPs [40].

Based on different ways of optimizing the SfM process, there are many open-source or commercial softwares that implement SfM for UAV modeling, such as OpenMVG, Contextcapture, Pix4d, and Photoscan. These softwares have been verified for a long time and multiple versions in terms of accuracy and reliability of UAV modeling. However, since their cores are based on the SfM algorithm, the errors in the bundle adjustment are generally consistent when considering the missing of GCPs. And an early study showed that the error in UAV modeling based on these softwares was only a few pixels [41], so we can confirm that the cumulative error mentioned above mainly comes from the SfM algorithm itself rather than other factors.

In this study, three kinds of GCPs are applied: RTK-based GCPs (RCPs), TLS-based GCPs (TCPs), and assumed GCPs (ACPs) [Fig. 3(a)]. Among them, RCPs can be acquired directly by RTK, which is easy to operate and has a relatively high accuracy [42]. The major drawback of this technology is that the workload in complex terrain is very high, and its accuracy may be reduced under the influence of weather, terrain, or tall trees; the most important thing is that it cannot be implemented under access restrictions [43]. In contrast, TCPs refer to the extraction of obvious feature points from the TLS point cloud after accurate alignment of all TLS data, such as the ends of electric poles, treetops, house corners, and protruding parts of boulders. TCPs can replace RCPs as the control points for UAVs under the condition that the point cloud density is large enough and the accuracy of multistation adjustment is high enough [30]. The advantage of this method is that it can overcome access restrictions and enable a large number of UAV control points or check points in places that are inaccessible to humans, which can significantly expand the

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distribution of GCPs. Even so, TCPs do not solve the problem of the lack of control points in the blind area. As for ACPs, they are artificially ACPs in the blind area, whose purpose is to provide external geometric conditions for the SfM algorithm regardless of errors. ACPs can be obtained by various methods, such as extracting the coordinates of feature objects from satellite images or previous contour maps [44]. Compared with RCPs or TCPs, ACPs obviously have a larger error, and once the error is too large, it will cause the accumulated error in the SfM algorithm to exceed the threshold and lead to the tearing of the UAV model [45].

C. Deformation Analysis Using ACPs in Blind Areas

Apparently, it is impossible to reconstruct an accurate UAV model by ACPs, but this does not prevent the use of ACPs for landslide deformation analysis. In this study, we provide a method for ground deformation analysis in blind areas using ACPs.

Landslides or surface displacements only change the terrain in their initial positions, stopping positions, and slide areas, while the surrounding ground features usually maintain the stability. Therefore, these stable features can be used as control points for model reconstruction, where it is most important to determine their coordinates as precisely as possible to avoid image tearing. As shown in Fig. 5(a), two different phases of the UAV model may be distorted along different directions during the initial aerial triangulation, which is usually unpredictable, so directly superimposing the two phases of UAV data for deformation analysis will result in great errors. Now, by adding several ACPs to the above UAV models, we can make these models distorted along to the same direction with the same magnitude in the SfM process. Then, the stable areas will exhibit complete overlap after image alignment, while the landslide areas will show significant image separation [Fig. 5(b) and (c)]. To avoid large errors in ACPs that cause model tearing, we suggest extracting ACPs from UAV data of one period and assigning them to other periods, or averaging the coordinates of ACPs after extracting them from multi-period UAV data. In addition, the most important point is that extracting ACP from the early UAV models allows the error distribution of ACP to obey the error distribution when the UAV model is far from GCPs. Therefore, such artificially selected control points will not be arbitrary and it is possible to be able to analyze their errors quantitatively.

Apparently, there are certain errors between the CD and the real displacement (RD), and we need to define and describe these errors as accurate as possible. Since the error of bundle adjustment increases with the number of registered aerial photographs, the error of the distorted model should be distance-dependent. Assuming that the model after using ACPs has been stretched with the polar axis centered on GCPs and assuming that there is a nonlinear functional relationship $\varepsilon(d)$ between the error of the model and the distance from real control points, then, the following relationship that exists between the RD and CD [Fig. 5(c)] is:

$$\mathrm{RD}_j = \mathrm{CD}_j - \left[\varepsilon\left(\boldsymbol{A}'_j\right) - \varepsilon\left(\boldsymbol{a}'_j\right)\right] \tag{2}$$

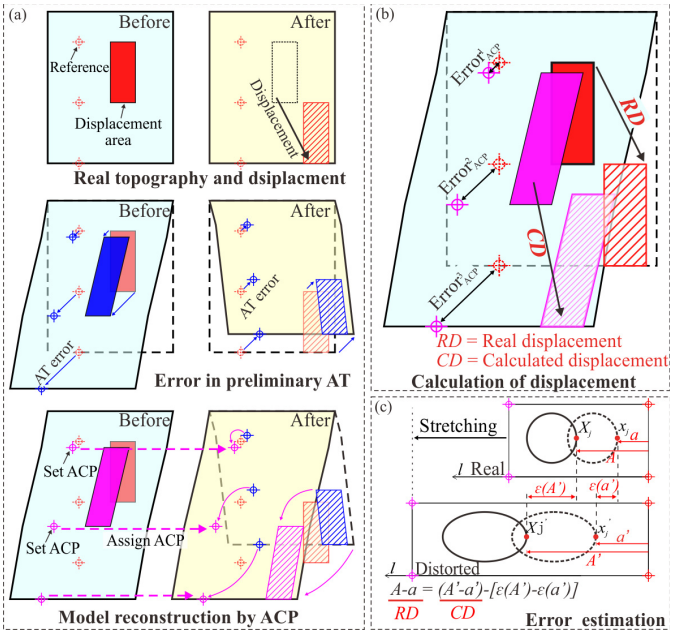


Fig. 5. Ground deformation analysis method for blind areas using ACPs: (a) distortion of assigned ACPs for reconstructing the UAV model; (b) deformation analysis of an aligned artificially distorted UAV model; and (c) relationship between RD and CD when the model is stretched (the point $\boldsymbol{x}_j$ moved to $\boldsymbol{X}_j$).

where $\boldsymbol{A}'_j$ is the polar diameter of point $\boldsymbol{x}'_j$ in the distorted UAV model from the real control point after deformation, and $\boldsymbol{a}'_j$ is the polar diameter of point $\boldsymbol{x}'_j$ in the distorted UAV model from the real control point before deformation.

Similarly, if the model undergoes compression along the polar diameter, then the relationship between RD and CD is

$$\mathrm{RD}_j = \mathrm{CD}_j + \left[\varepsilon\left(\boldsymbol{A}'_j\right) - \varepsilon\left(\boldsymbol{a}'_j\right)\right]. \tag{3}$$

Therefore, if the distribution of function $\varepsilon(d)$ can be determined, then the relationship between RD and CD can be quantitatively analyzed.

D. Calculation of Ground Displacement

To analyze landslide deformation, it is possible to consider using 3-D point clouds, digital surface models (DSMs), or orthophotos [2], [46]–[50]. Since the H02 landslide is slow sliding along the slope, if the 3-D deformation analysis based on the 3-D point cloud, such as the C2C [51], C2M [52], or M3C2 [53], is used for the deformation analysis, it is easy to mismatch the feature points before and after the deformation, thus resulting in errors in the deformation analysis [30]. In contrast, the horizontal displacement of the landslide can be calculated by extracting the feature points before and after deformation on the orthophoto with RGB values [54]. Meanwhile, the horizontal displacement is synchronized with the actual displacement when the landslide slides along the slope, so we consider using the orthophoto for surface deformation analysis.

Surface displacement analysis using the difference of orthophotos (DOO) is a relatively simple method in which the researcher only needs to find the position of the same feature point on two maps separately, such as a stone, the

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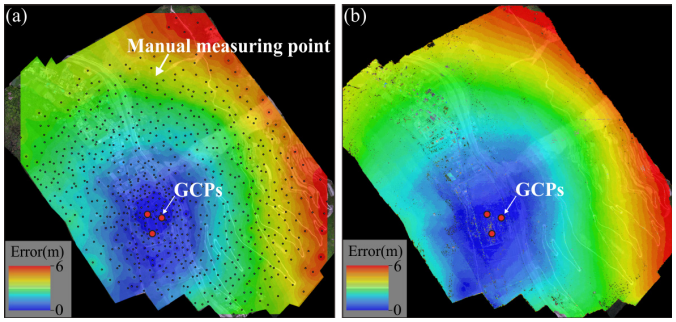


Fig. 6. Comparison of manual DOO and COSI-Corr-based DOO: (a) manual DOO of two orthophotos using 770 feature points and (b) correlation analysis of the orthophotos based on COSI-Corr.

corner of a house, or a treetop, and then use the line between the two points to represent the displacement vector [55], [56]. However, it would be very time-consuming to match all the feature points manually. In this study, we use co-registration of optically sensed images and correlation (COSI-Corr) [57] for the automatic calculation of the relative displacement of two UAV orthophotos. The "Correlation" function in this software is an image recognition program based on a correlation algorithm. In this function, the correlation between two strictly aligned orthophotos can be analyzed by setting a certain size of search window that traverses the whole orthophoto in a certain manner to obtain the correlation values, and then, the overall correlation of the images can be analyzed statistically to finally calculate the relative displacement between the two images [58]. In addition, if the lushness and color of vegetation change significantly between the two orthophotos due to seasonal changes, it is recommended to use the hill shadow maps generated by DSM instead of the original RGB images for correlation analysis [59].

As shown in Fig. 6, we manually extracted 770 feature points in a DOO process and compared it with COSI-Corr. The result shows a high consistency between the two methods, but the time required to complete the DOO using the manual analysis method is ten times longer than that of COSI-Corr.

III. Results

A. Error Evaluations

For the extraction of ACPs from a distorted UAV model and the calculation of the surface deformation in a blind area, it is necessary to quantify the accuracy of the UAV model outside the influence range of the GCPs and provide a way to estimate the errors.

As shown in Fig. 7, we clipped an area within the visual range inside the study area where no significant deformation occurred and studied how the UAV model was distorted outside the influence range of the real control points. The vast part of this region can be directly observed by TLS, so we can add or subtract TCPs to the majority of visible areas. We used two orthophotos taken in winter (Ortho. A on December 6, 2020) and summer (Ortho. B on May 10, 2021), each considering six distributions of three TCPs in terms of vegetation, height difference, and spacing of control points. Then, we used the orthophotos generated from each scenario

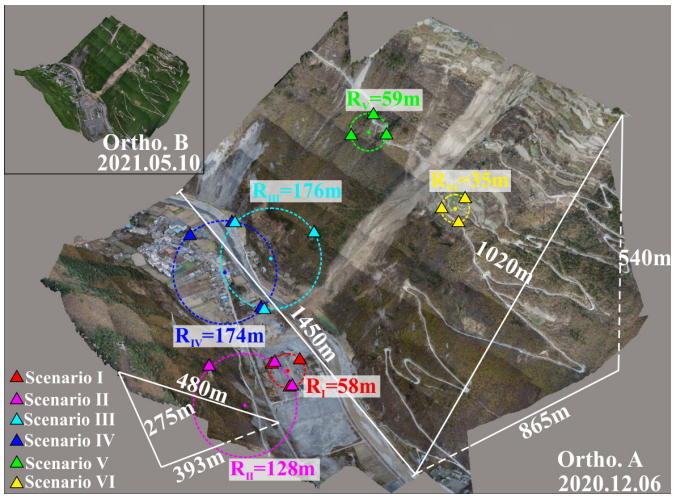


Fig. 7. Six scenarios of control point arrangement in the form of three TCPs for each scenario, where R indicates the radius of the circle enclosed by the three TCPs. Among them, scenarios 2 and 3 have two TCPs within the river valley and one TCP on the hillside, scenarios 1 and 4 have all three TCPs, and scenarios 5 and 6 have TCPs within the river valley and on the hillside.

to perform a DOO analysis with the real orthophotos. The results of the DOO are shown in Fig. 8.

Apparently, within the influence range of the control point or with a certain distance outside the control point, the accuracy of the UAV model is acceptable, but as the distance gradually increases, the accuracy of the UAV model rapidly decreases. However, the reduction in accuracy is not linearly related to the distance to the control point, and the direction of the error is mostly rotationally distributed around the GCPs, which is random and unpredictable according to the white arrows.

For further quantitative analysis, we performed a statistical analysis using the error values in these 12 DOO results as the y-axis and the distance of each cell from the center of the control point as the x-axis (Fig. 9). In addition, we calculated the mean value of the error and highlighted the distance corresponding to an average error equal to ten pixels.

According to the results, the UAV model error tends to rise smoothly and slowly in the area of 0–300 m from the nearest control points. When the distance is between 400 and 600 m, the error may then continue to rise or may show a small decrease; that is, there may be more significant uncertainty in the error of the UAV model at this distance. As the distance increases, the error rises again after the distance exceeds 800 m. In general, the accuracy of the UAV model is stable and predictable up to 300 m from the nearest control point, while it varies considerably over 600 m. In addition, UAV accuracy for scenarios 5 and 6 is relatively lower than that for the other cases, especially scenario 6. This may be because their control points are located on hillsides with steep slopes and surrounded by a large amount of vegetation, which increases the cumulative error in the SfM algorithm.

Furthermore, we summarized the mean value and standard deviation of the error in all scenarios, as shown in Fig. 10. Then, we fitted the values using the allometricl equation and acquired the function of $\varepsilon(d)$

$$\varepsilon(d) = (2.64 \times 10^{-4} \pm 6.65 \times 10^{-5}) * d^{1.76 \pm 0.04}. \quad (4)$$

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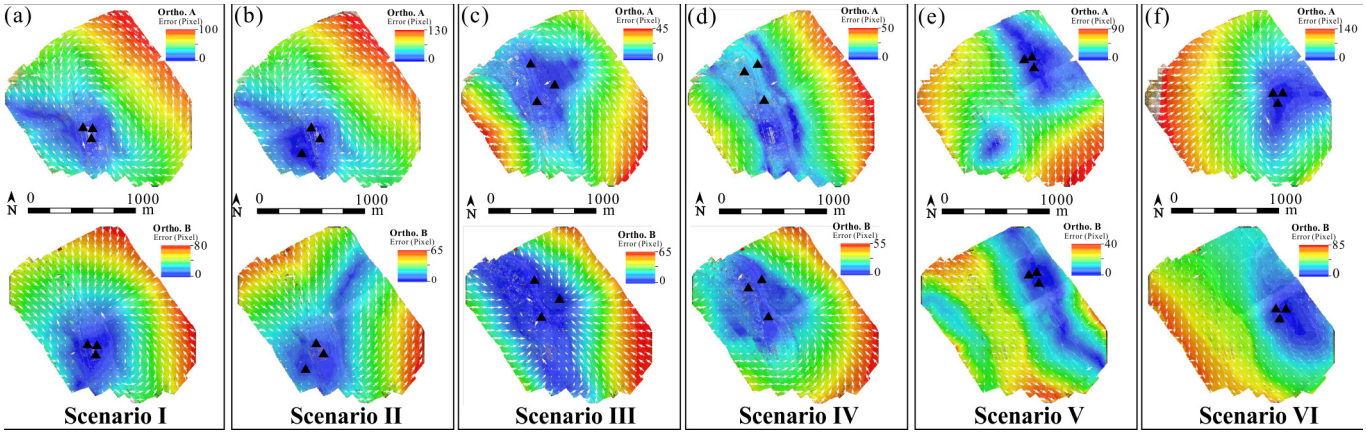


Fig. 8. Error distribution for six scenarios of control point arrangement, the results of which are obtained by comparing each scenario with the real orthophoto (the white arrows indicate the distortion direction). (a) and (d) Three TCPs within the river valley. (b) and (c) Two TCPs within the river valley and one TCP on the hillside. (e) and (f) Three TCPs on the hillside. (One pixel equals 5 cm.)

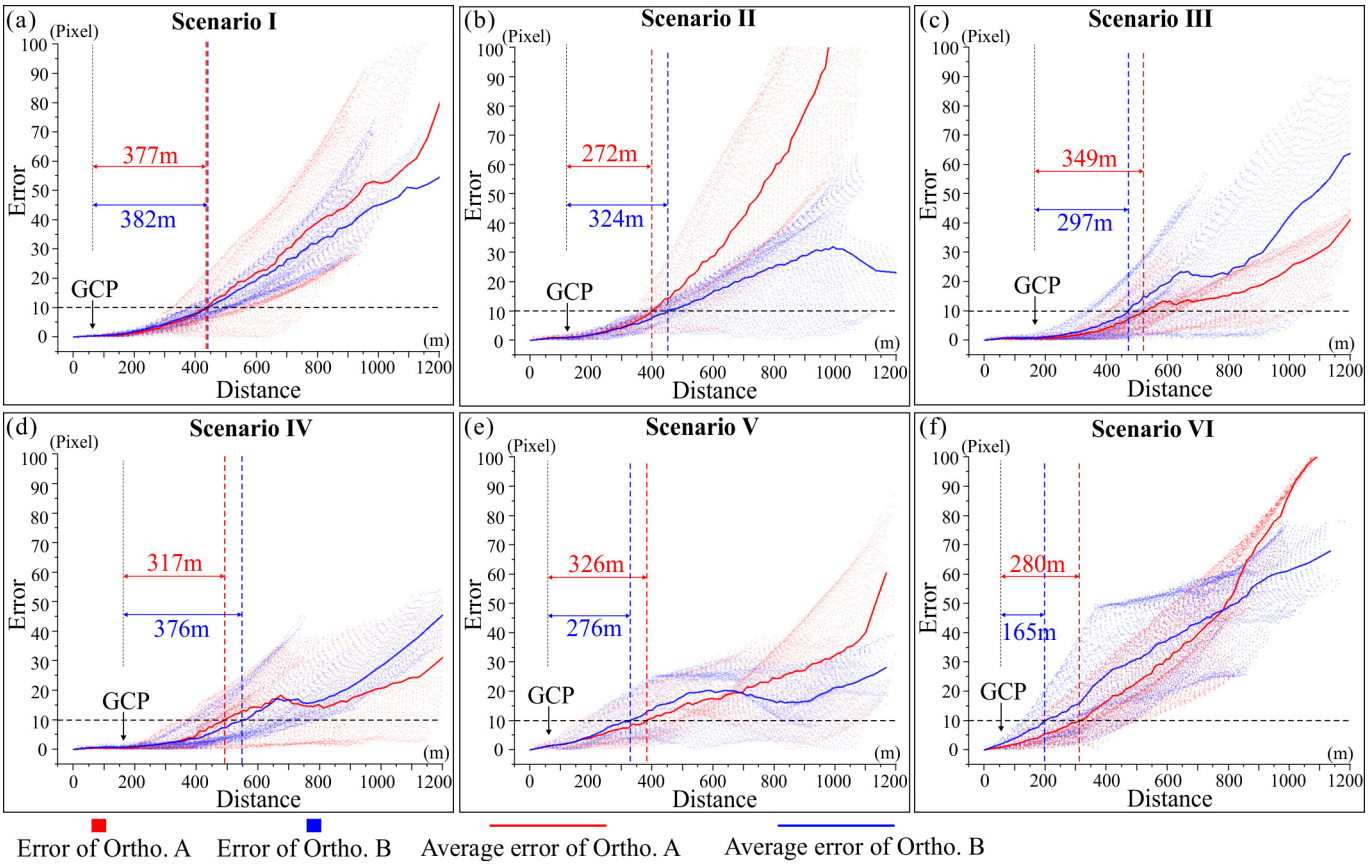


Fig. 9. Statistical analysis of error distribution in Fig. 8, where the dots correspond to the error of each pixel, and the solid lines refer to the average error. (a) and (d) Three TCPs within the river valley. (b) and (c) Two TCPs within the river valley and one TCP on the hillside. (e) and (f) Three TCPs on the hillside. (One pixel equals 5 cm.)

As shown in Fig. 11(a), the distortion of the UAV model is usually a rotational twist centered on the control point, which is a random and very unpredictable process. Thus, it is difficult to define the quantitative relationship between CD, RD, and $\varepsilon(d)$. However, it is worth noting that a relationship exists

$$\mathrm{CD}_j = \boldsymbol{A}'_j - \boldsymbol{a}'_j. \tag{5}$$

If the UAV model stretches or compresses in the direction of the polar axis of the control point [Fig. 11(b) and (c)], (2) and (3) can be rewritten as follows:

Stretching:

$$\mathrm{RD}_j^{\min} = \mathrm{CD}_j - \left[\varepsilon\left(\boldsymbol{a}'_j + \mathrm{CD}_j\right) - \varepsilon\left(\boldsymbol{a}'_j\right)\right]. \tag{6}$$

Compressing:

$$\mathrm{RD}_j^{\max} = \mathrm{CD}_j + \left[\varepsilon\left(\boldsymbol{a}'_j + \mathrm{CD}_j\right) - \varepsilon\left(\boldsymbol{a}'_j\right)\right]. \tag{7}$$

When the UAV model undergoes rotational distortion, RD should satisfy the following relationship according to the geometric relationship:

$$\mathrm{RD}_j^{\min} \le \mathrm{RD}_j \le \mathrm{RD}_j^{\max}. \tag{8}$$

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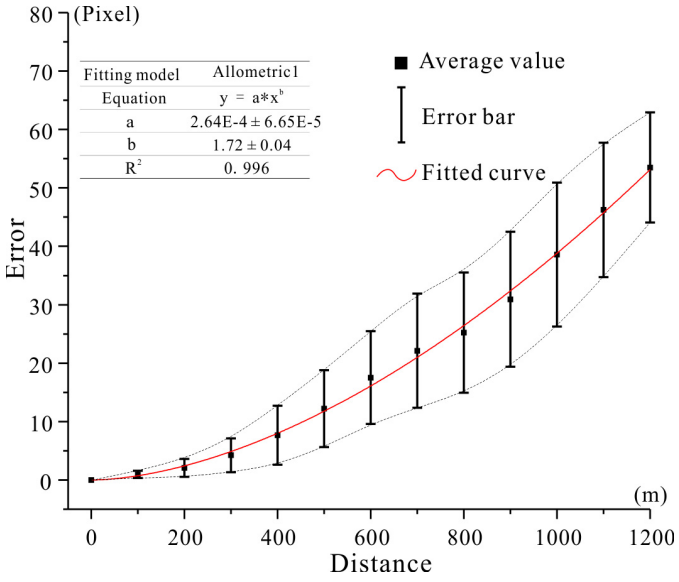


Fig. 10. Total mean and standard deviation of all six scenarios and the fitted allometricl function of $\varepsilon(d)$. (One pixel equals 5 cm.)

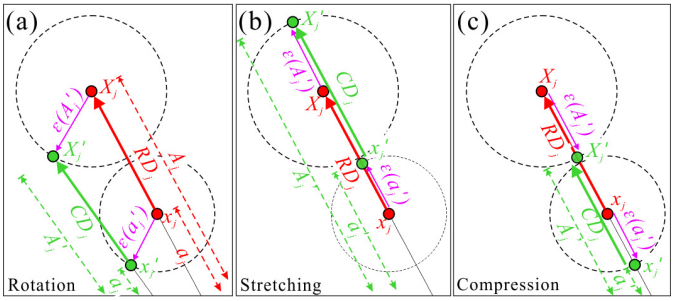


Fig. 11. Relationship between CD, RD, and $\varepsilon(d)$ in: (a) rotation; (b) stretching; and (c) compression.

At this point, only $\boldsymbol{a}'_j$ and CD_j are needed to calculate the values of RD_j, and these two values can be extracted directly from the deformation analysis results. According to (6)–(8), we exemplify some of the calculation results that can be used as a reference to determine the errors between RD_j and CD_j, where the maximum value corresponds to compression and the minimum value corresponds to stretching (Table II). The values represent the deviation between RD_j and CD_j when a ground displacement of CD_j occurs at a distance of $\boldsymbol{a}'_j$ from the nearest control point.

B. Deformation Analysis Error Using ACPs

With the quantitative results for $\varepsilon(d)$, we analyze the feasibility and error of the deformation analysis method proposed in Section II-C. This step is based on scenario I in Fig. 8(a), in which two ACPs (#1 and #2) are added to simulate surface deformation measurements in blind areas. As shown in Fig. 12, we added ACP #1 and ACP #2 at distances of approximately 500 and 800 m from the nearest control point, respectively, and set up three scenarios, corresponding to the settings of ACP #1, ACP #2, and ACPs #1 and #2. The two ACPs are extracted from Ortho. A and assigned to Ortho. B. After the reconstruction, we superimpose the two reconstructed orthophotos and analyze the deviation between them. The

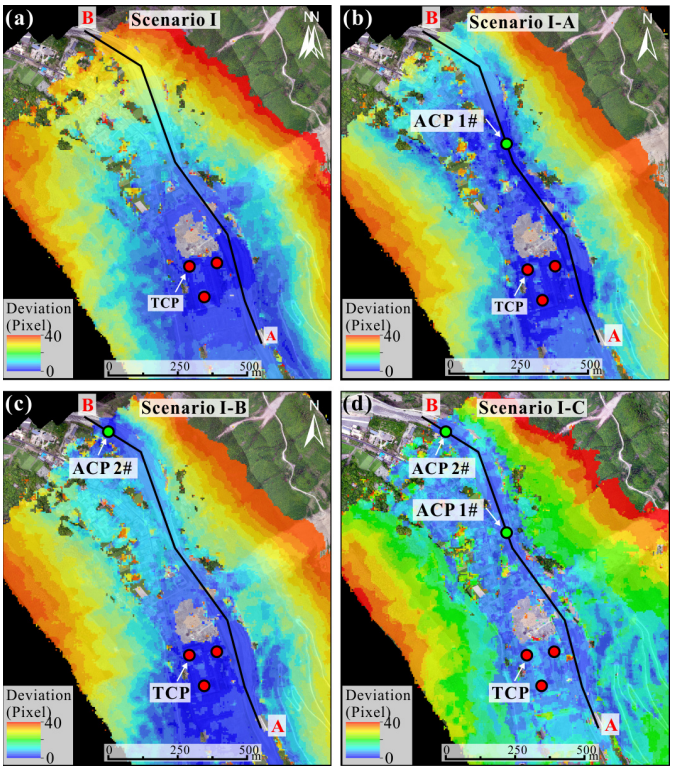


Fig. 12. Deviation distribution of Ortho. A and Ortho. B under different control point distribution scenarios: (a) scenario I in Fig. 8(a); (b) ACP #1 is added approximately 500 m from the TCPs; (c) ACP #2 is added approximately 800 m from the TCPs; and (d) ACPs #1 and #2 are added at the same time. (One pixel equals 5 cm.)

deviation distribution and details of reconstruction with ACPs are shown in Figs. 12 and 13. According to the results, the ACPs force the two orthophotos, which originally had obvious deviations, to align. As shown in Fig. 13, the deviations of the reconstructed model and real model near ACP #1 and ACP #2 are approximately 38 pixels with 800 m and 12 pixels with 400 m, respectively, which coincide with the error distribution of $\varepsilon(d)$ in Fig. 10.

For further quantitative analysis, a profile line along the river channel of XM township was selected, and its deviation from the four scenarios was generated (Fig. 14). Apparently, when no ACP is used (the red dots), as long as the distance from the real GCP reaches approximately 200 m, the deviation between the two orthophotos will exceed five pixels. In contrast, after adding an ACP at a distance of approximately 500 m from the real control point (the green dots), the deviation within the range covered by ACP #1 and real GCPs can be greatly reduced, but as the distance continues to increase, the deviation will rapidly rise. In addition, the distance of an ACP should not be too far from the real GCP; otherwise, it can only improve the accuracy near the ACP and cannot effectively reduce the error in the area between the real control points and the ACP (the blue dots of ACP #2).

Next, we need to analyze the errors of surface deformation analysis using the ACP-reconstructed model. However, the study area we selected at this point is actually without obvious surface displacement, and the vast majority of the areas of the two orthophotos are completely overlapping, so it is

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TABLE II
ERROR BETWEEN RD_j AND CD_j

Error proportion: 0~0.10% 0.10%~0.15% 0.15%~0.20% 0.20%~0.25% 0.25%~0.30%

RD_j (m) \ a_j' (m)	100	200	300	400	500	600	700	800
5	0.02 ± 0.02	0.06 ± 0.01	0.15±0.07	0.18 ± 0.08	0.21 ± 0.10	0.25 ± 0.12	0.28 ± 0.13	0.31 ± 0.15
10	0.12 ± 0.05	0.16 ± 0.07	0.29±0.13	0.37 ± 0.17	0.43 ± 0.21	0.50 ± 0.24	0.56 ± 0.27	0.62 ± 0.30
15	0.19 ± 0.08	0.31 ± 0.14	0.44±0.20	0.55 ± 0.26	0.65 ± 0.31	0.74 ± 0.36	0.84 ± 0.41	0.92 ± 0.46
20	0.24 ± 0.10	0.57 ± 0.26	0.58±0.27	0.73 ± 0.35	0.86 ± 0.42	0.99 ± 0.48	1.11 ± 0.55	1.23 ± 0.61
30	0.35 ± 0.15	0.85 ± 0.39	0.87±0.40	1.08 ± 0.51	1.29 ± 0.62	1.48 ± 0.72	1.66 ± 0.82	1.84 ± 0.92
40	0.45 ± 0.19	1.11 ± 0.52	1.14±0.53	1.43 ± 0.68	1.71 ± 0.82	1.96 ± 0.96	2.21 ± 1.09	2.44 ± 1.22
50	0.54 ± 0.23	1.37 ± 0.64	1.41±0.66	1.78 ± 0.84	2.12 ± 1.02	2.44 ± 1.19	2.75 ± 1.35	3.04 ± 1.51
60	0.61 ± 0.26	1.63 ± 0.75	1.67±0.78	2.11 ± 1.00	2.52 ± 1.21	2.91 ± 1.42	3.28 ± 1.62	3.63 ± 1.81
80	0.72 ± 0.30	2.11 ± 0.97	2.17±1.00	2.76 ± 1.31	3.31 ± 1.59	3.83 ± 1.87	4.32 ± 2.13	4.80 ± 2.39
100	0.77 ± 0.32	2.55 ± 1.18	2.63±1.22	3.38 ± 1.60	4.07 ± 1.96	4.72 ± 2.30	5.35 ± 2.63	5.94 ± 2.95

The unit of the error between RD_j and CD_j is a pixel, and one pixel equals 5 cm.

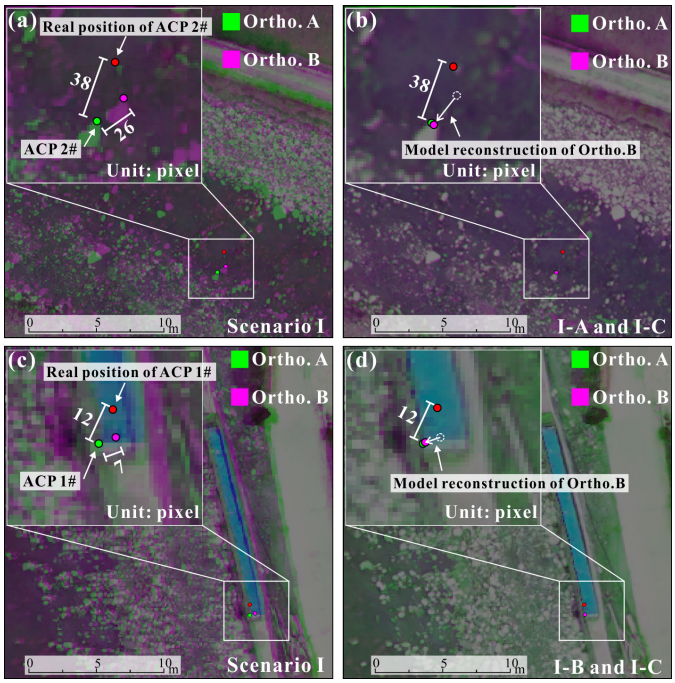


Fig. 13. Details of the model reconstruction based on ACPs: (a) and (b) reconstruction based on ACP #2 and (c) and (d) reconstruction based on ACP #1.

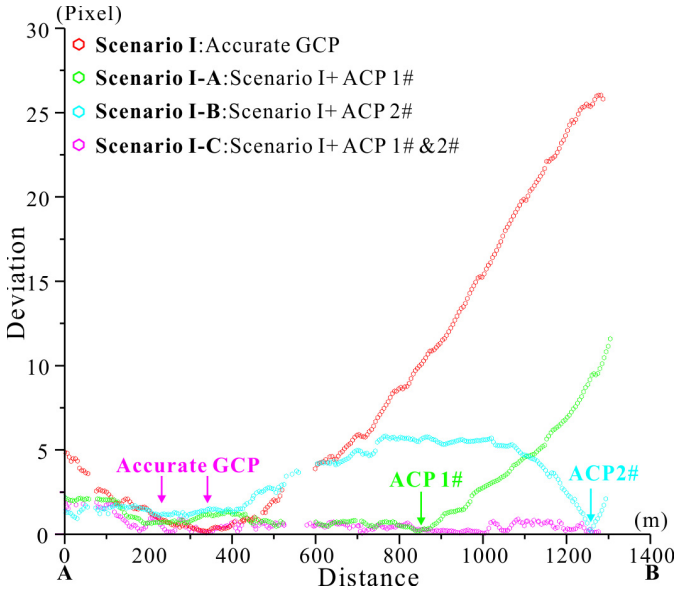


Fig. 14. Deviation between Ortho. A and Ortho. B along the profile in Fig. 9; one pixel equals 5 cm.

impossible to extract RD and CD by finding the objects where displacement occurs. Therefore, we considered an alternative approach. The essence of surface displacement analysis with DOO is to find a set of corresponding points from two strictly aligned orthophotos. Therefore, in the area where no surface deformation occurs, we can find any two points to form a point pair and assume that one of them moves to the other, i.e., a virtual displacement. Based on this idea, we selected six initial points (six obvious stones in the river channel) near the profile line in Fig. 12, which are approximately 300, 400, 500, 600, 700, and 800 m away from the nearest control point (i.e., $\boldsymbol{a}_j'$), and then assumed that they were displaced 60 m along the upstream direction of the river (the northwest). After that, we obtained the theoretical errors between CD and RD for the six point pairs from Table II and compared them with the actual deviation between the reconstructed model and real model (see Table III). The results show that all actual deviations fall within the range of the theoretical errors, thus proving that the errors between CD and RD can be effectively estimated by (6)–(8) when ACPs are used for surface deformation analysis. The value of error is small in proportion to the total displacement, so CD can be used to represent RD to indicate the surface displacement in this situation.

C. Case Application

We applied the above method to the deformation monitoring of the H02 landslide. The two phases of UAV data were collected on December 6, 2020 (Ortho. A) and May 10, 2021

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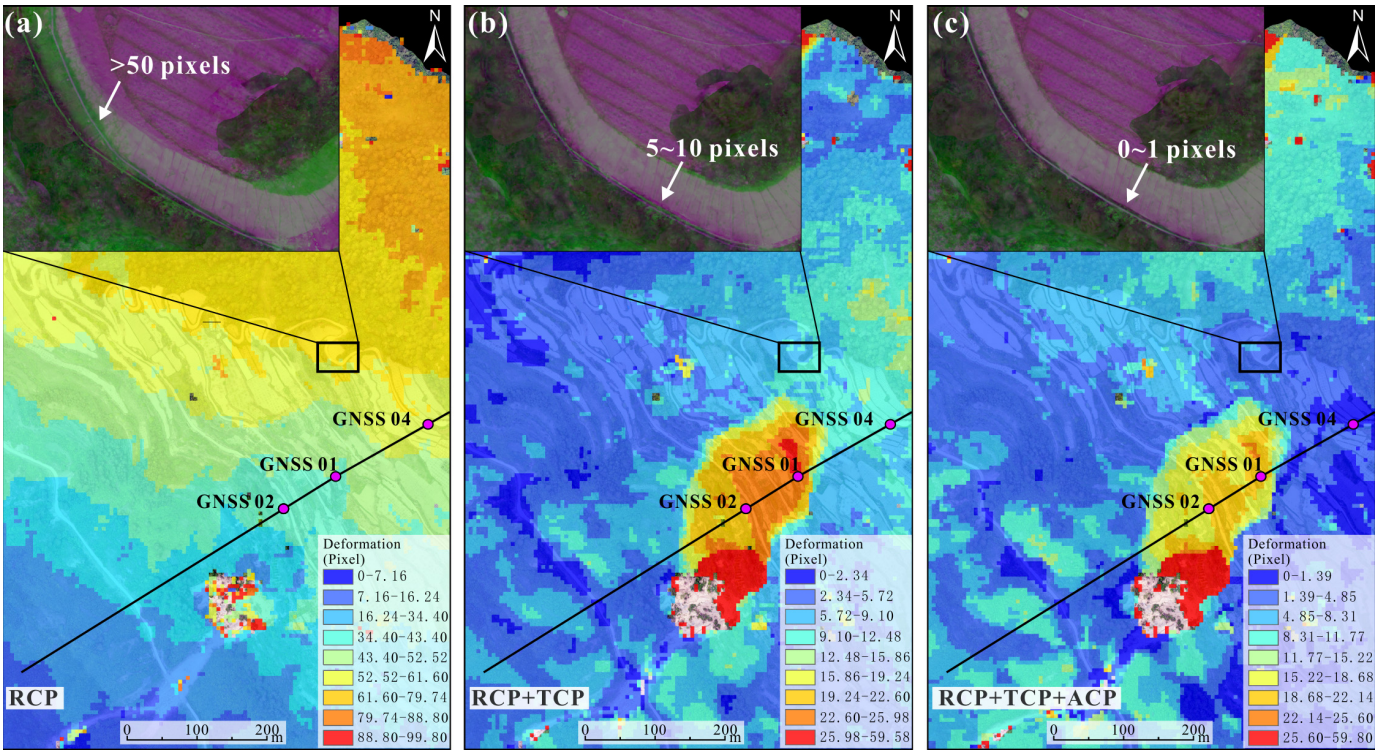


Fig. 15. Deformation analyses of the H02 landslide using DOO between Ortho. A and Ortho. B: (a) deformation analysis based on RCPs along the river valley; (b) deformation analysis after adding TCPs near the sight boundary; and (c) deformation analysis based on ACP-reconstructed orthophotos.

(Ortho. B) with a resolution of 5 cm. The emergency road was built approximately one month after our first UAV data collection, and then, the local government commissioned a third-party company to install GNSS in QL village, which is solar-powered and can remotely transmit data to the online monitoring platform.

For comparative analysis, we performed a total of three scenarios of H02 landslide deformation analysis, as shown in Fig. 2.

1) Ten RCPs were arranged in the accessible area by RTK.
2) Based on the RCPs, five TCPs were added near the sight boundary using TLS, and the closest TCP was approximately 100 m from the front edge of H02 and approximately 400 m from its back edge.
3) Three ACPs were extracted from Ortho. A generated by RCPs + TCPs and used to reconstruct both Ortho. A and Ortho. B. Each ACP was approximately 500 m from the nearest TCP.

Fig. 15 shows the results of the landslide deformation analysis of H02, and the profile in Fig. 15 is shown in Fig. 16. When only RCPs are used, the surface deformation appears to be obviously wrong, and even the body shape of the landslide cannot be accurately judged [Fig. 15(a)]. In fact, this highlights a current situation in the landslide monitoring field, in which it is almost impossible for UAV photogrammetry to be used for surface deformation monitoring under access restrictions.

As shown in Fig. 15(b), the addition of TLS can greatly improve the effectiveness of landslide monitoring where access restrictions exist. Although the location of H02 is outside the influence range of the TCPs, the accuracy of the landslide area

TABLE III
ERROR OF GROUND DISPLACEMENT ANALYSIS

a_i' (m)	RD (m)	CD (m)	Actual error (meters) (pixels)	Theoretical error (pixels)	Proportion of error (%)
300	60.380	60.425	(0.045) (0.89)	1.67 ± 0.78	0.07%
400	60.621	60.687	(0.066) (1.33)	2.11 ± 1.00	0.11%
500	59.662	59.738	(0.076) (1.51)	2.52 ± 1.21	0.13%
600	60.207	60.319	(0.112) (2.22)	2.91 ± 1.42	0.18%
700	61.704	61.786	(0.082) (1.63)	3.28 ± 1.62	0.13%
800	61.220	61.468	(0.248) (4.96)	3.63 ± 1.81	0.41%

The theoretical error can be obtained in Table II or calculated by Eqs. 6-8. (One pixel equals 5 cm.)

is greatly improved due to the close distance to the TCPs, and we can not only see the boundary of the landslide body but also roughly estimate its deformation magnitude. However, since the area between the RCPs and TCPs fails to contain the back edge of the landslide, the location of the landslide boundary cannot be accurately defined at this stage. From the subgraph in Fig. 15(b), there is still a deviation of 5–10 pixels in the two phases of the UAV orthophoto at the back edge of the landslide. Without other information, it is difficult to determine whether this deviation is surface displacement or model error, which creates great confusion in determining the true extent or scale of landslides. Especially for the safety monitoring of landslide emergency management, if we cannot indicate whether a certain area is deformed or not, then reasonable and correct measures will not be available, and the safety of personnel and equipment on the site cannot be guaranteed.

As shown in Fig. 15(c), with the addition of ACPs, the accuracy of the deformation analysis near the back edge of

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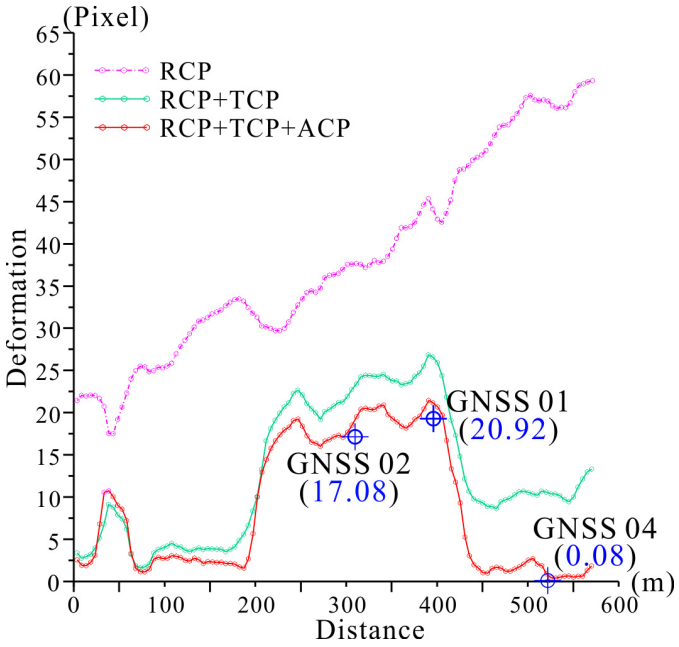


Fig. 16. Profiles of deformation analysis results. (One pixel equals 5 cm.)

TABLE IV
DISPLACEMENT ANALYSIS OF GNSS 01 AND GNSS 02

Position	Displacement of GNSS (m)	Distance to real GCP (m)	CD (m)	Theoretical error (pixels)
GNSS 01	1.046	360	1.116	0.03 ± 0.01
GNSS 02	0.854	290	0.937	0.02 ± 0.01

One pixel equals 5 cm.

H02 was effectively increased, and the error was reduced from approximately 10 pixels to approximately 1 pixel. In addition, we labeled the monitoring data of GNSS, whose monitoring results are in good agreement with CD. The details are shown in Table IV. It is worth noting that the beginning of GNSS monitoring was 26 days after the acquisition of Ortho. A, so the GNSS monitoring value should be slightly lower than the real value, which is consistent with the results in Fig. 16. According to (6)–(8), we deduce the theoretical errors of 0.03 ± 0.01 pixels and 0.02 ± 0.01 pixels for the positions of GNSS 01 and GNSS 02, respectively, so it is possible to consider using CD to represent RD with such a small error.

IV. DISCUSSION

The premise of using the UAV photogrammetry for surface deformation analysis is to ensure that the multiple periods of data can be accurately aligned. In general, the traditional solution is based on geodesic coordinates, which first completely restore the real position and shape of the UAV model in 3-D space, and then align the models and calculate their differences [60], [61]. However, this approach makes it impossible to reconstruct a high-precision UAV model in the absence of control points in blind areas. In contrast, the method proposed in this article allows the UAV model to maintain a certain error in invisible areas, then uses ACPs to generate artificially distorted but accurately aligned terrain data, and finally achieves highly accurate surface deformation analysis based on a quantitative error formula.

In (6) and (7), the errors between RD and CD are mainly affected by three factors: the error distribution formula outside the control point [$\varepsilon(d)$], the distance between the displaced point and the nearest control point (a'_j), and the CD of the landslide in the reconstructed models (CD_j). Among them, the $\varepsilon(d)$ proposed in this article is statistically derived from combining 12 scenarios of control point distribution in the study area, and its parameters mainly characterize the influence of topographic factors such as slope, elevation difference, and vegetation on model accuracy. For $\varepsilon(d)$ under different terrain conditions, the specific parameters may be different. Therefore, for different study areas, multiple sets of GCPs for different scenarios can be considered to calculate the corresponding $\varepsilon(d)$. However, as mentioned earlier, it is very difficult to set up manual control points in areas with complex terrain conditions, so TLS can play an important role in obtaining accurate $\varepsilon(d)$ in such cases.

Since the error of surface displacement generated by using ACP is only less than 1% of the total displacement, is it possible to replace all real control points with ACP in the actual monitoring process, so that the UAV does not need to do any preliminary work when monitoring landslide deformation? We address this question and argue that the selection of ACPs is restricted and can only be selected in stable areas where no surface deformation has occurred. Due to the existence of the blind area, there is no reference for ACP selection at the initial stage, so the point selected as the ACP may have been displaced. As shown in Fig. 17(a), we simulated a scenario with an incorrect example by replacing an ACP outside the landslide area in Fig. 2 with an ACP inside the landslide area. Fig. 17(b) shows the distribution of surface deformation under this scenario, and Fig. 17(c) and (d) shows two typical profiles. As seen from the results, once the ACP is selected in an area where deformation has occurred, the results of the surface deformation analysis will be significantly erroneous.

In SfM algorithm, the SIFT-like algorithms have been widely used because they not only have good adaptability to rotation, scale scaling, and luminance changes, but also maintain a certain degree of stability to viewpoint changes, affine transformations, and noises. However, it is relatively slow in computation because it needs to traverse all potential feature points in the photographs. And for aerial images taken under certain extreme conditions, these methods also have problems such as numerous outliers and insufficient key points. For example, for each scenario designed in this article, two computers (CPU: i7-11700KF, GPU: RTX3060, software: UAV Manager provided by Feimarobotics, Shenzhen, China [32]) are required to work simultaneously for about 1 h to complete the SfM process and 31 h to reconstruct the UAV model, respectively. Therefore, when applied to scenarios where the number of aerial images is high, it is not possible to effectively respond to time-sensitive tasks such as emergency monitoring, emergency rescue, and emergency risk avoidance. In contrast, with the development of deep learning, algorithms such as convolutional neural networks (CNNs) [63] and graph convolutional networks (GCNs) [64] have achieved

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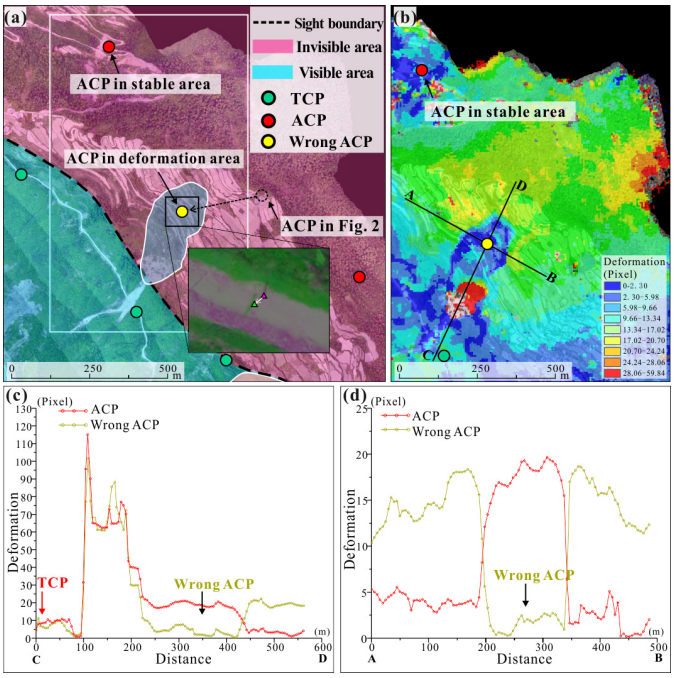


Fig. 17. Setting ACPs in the landslide area: (a) layout of incorrect ACPs; (b) ground deformation analysis based on incorrect ACPs; and (c) and (d) two profiles indicating the details of the incorrect ACPs. (One pixel equals 5 cm.)

remarkable results in the field of feature extraction, image classification, and image registration, and seem to outperform SIFT-like algorithms in many aspects such as speed, accuracy, robustness, and matching tasks [65], [66]. Compared with traditional methods, deep learning transforms the construction of hand-craft detector into the building of more complex and information-rich networks and, thus, has a broader development space and application prospects under more complex and challenging image registration tasks [20], [67].

V. Conclusion

This article proposed a fusion method using TLS and UAV photogrammetry for landslide deformation monitoring in complex terrain conditions. The core of the method is to use TLS to overcome the limitation of terrain restrictions on the deployment of GCPs for UAVs and then to use ACPs to reconstruct the UAV model for ground deformation analysis in blind areas. Based on a statistical analysis of the error distribution for 12 scenarios, we proposed an allometricl formula $\varepsilon(d)$ to predict the deviation between CD and RD when measuring surface deformation based on ACP-reconstructed orthophotos. To validate the method, we selected a small stable area of the study area to test and found that the errors in CD and RD were as expected in $\varepsilon(d)$. Finally, we applied the method to the H02 landslide in QL village and successfully realized landslide displacement monitoring under the conditions of access and area limitations. The calculated results were compared with the monitoring data obtained by GNSS, and we found that the proposed method can indeed solve the problem of surface deformation monitoring in complex terrain, and the error is acceptable. UAV modeling based on traditional detectors still has problems such as slow data processing speed when applied to time-sensitive emergency tasks, so the field of deep learning, which is more promising in terms of efficiency, robustness, and speed, will have a wide range of prospects in the application of UAVs to geological disaster prevention and mitigation [68].

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